CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 5



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Recap of Previous Lecture



Topic Cache Mapping

Topic Direct Mapping

Topic Tag

Topics to be Covered



- Topic: Cache Mapping
- Topic: Direct Mapping
- Topic: Tag
- Topic: Tag Directory



Topic : Direct Mapping

Tag = 1 digit $10^1 = 10$

Cache	0	1	2	3	4	5	6	7	8	9
Main Memory	00	01	02	03	04	05	06	07	08	09
	10	11	12	13	14	15	16	17	18	19
	20	21	22	23	24	25	26	27	28	29
	30	31	32	33	34	35	36	37	38	39
	:	:	:	:	:	:	:	:	:	:
	90	91	92	93	94	95	96	97	98	99

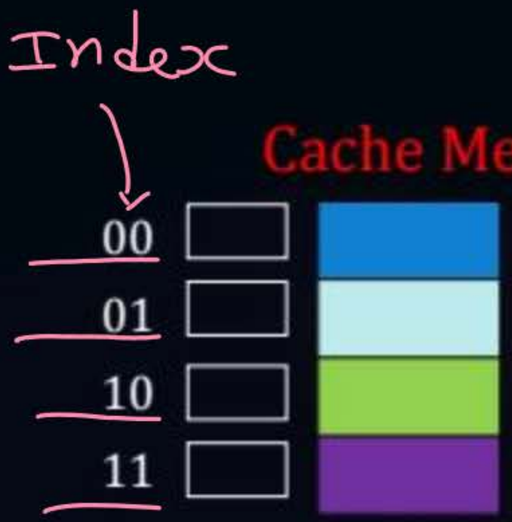


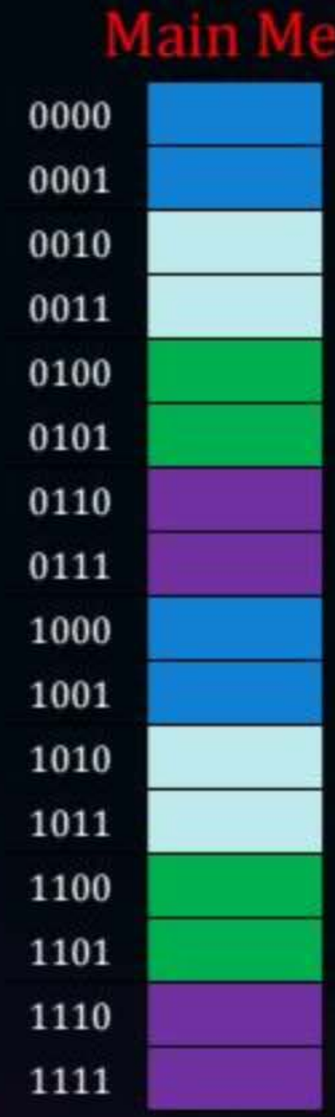
Topic : Direct Mapping

- Blocks in cache = 4 (00-11)
- Blocks in Main memory = 8 (000-111)
- Block Size = 2 Bytes = 2^1 B
- Size of Cache memory = 8 B
- Size of Main memory = 16 B
- Size of Main memory address = 4 bits



Topic : Direct Mapping





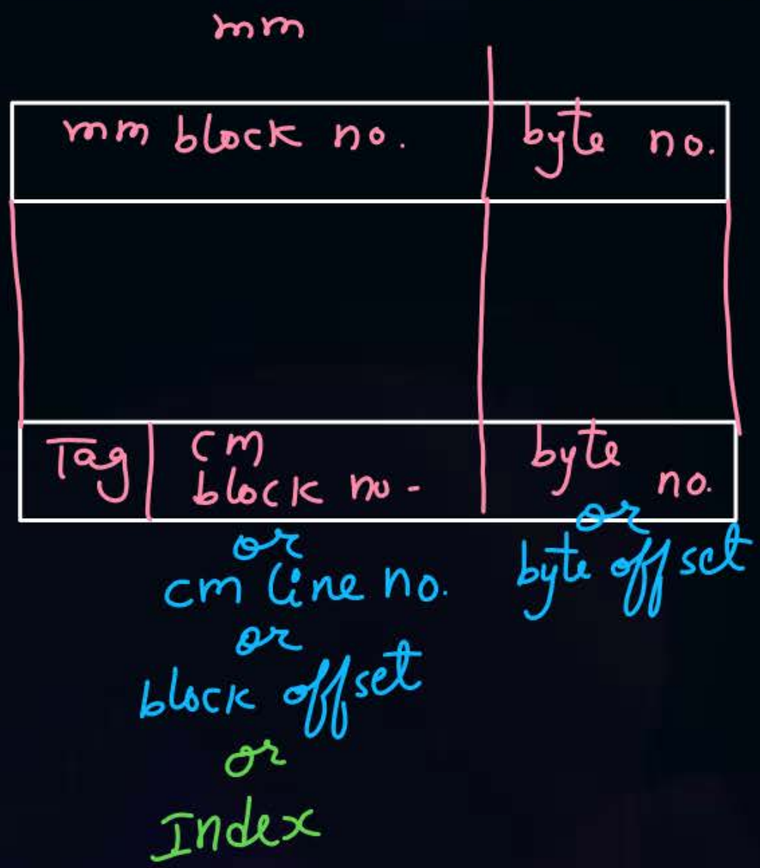
byte offset

↓

needed to identify a byte within a block.



Topic : Direct Mapping



$$\text{no. of bits for byte offset} = \log_2(\text{block size})$$

$$\text{no. of blocks in cache} = \frac{\text{cache size}}{\text{block size}}$$

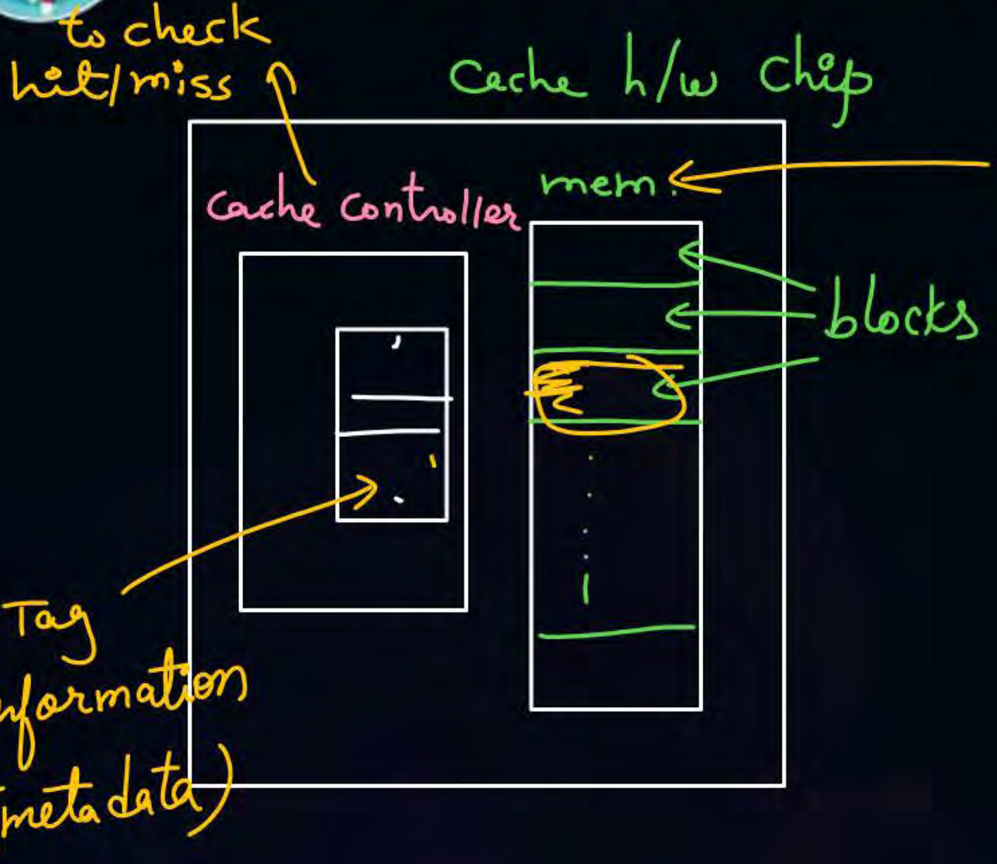


Topic : Indexing in Direct Mapping

cm block no. is index in direct mapping



Topic : Cache Controller



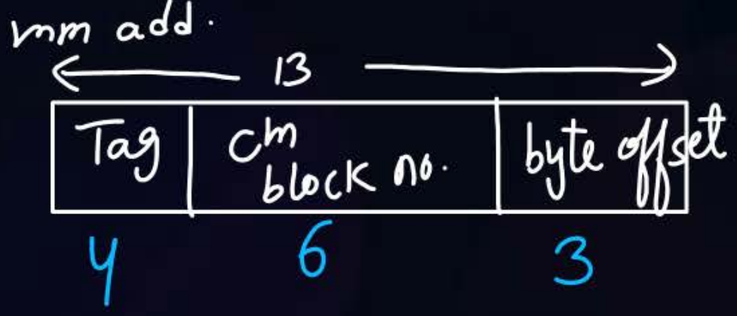
Tag directory size or metadata size = no. of blocks in cache * Tag bits

[NAT]

$2^3 B \Rightarrow$ block offset = 3 bits

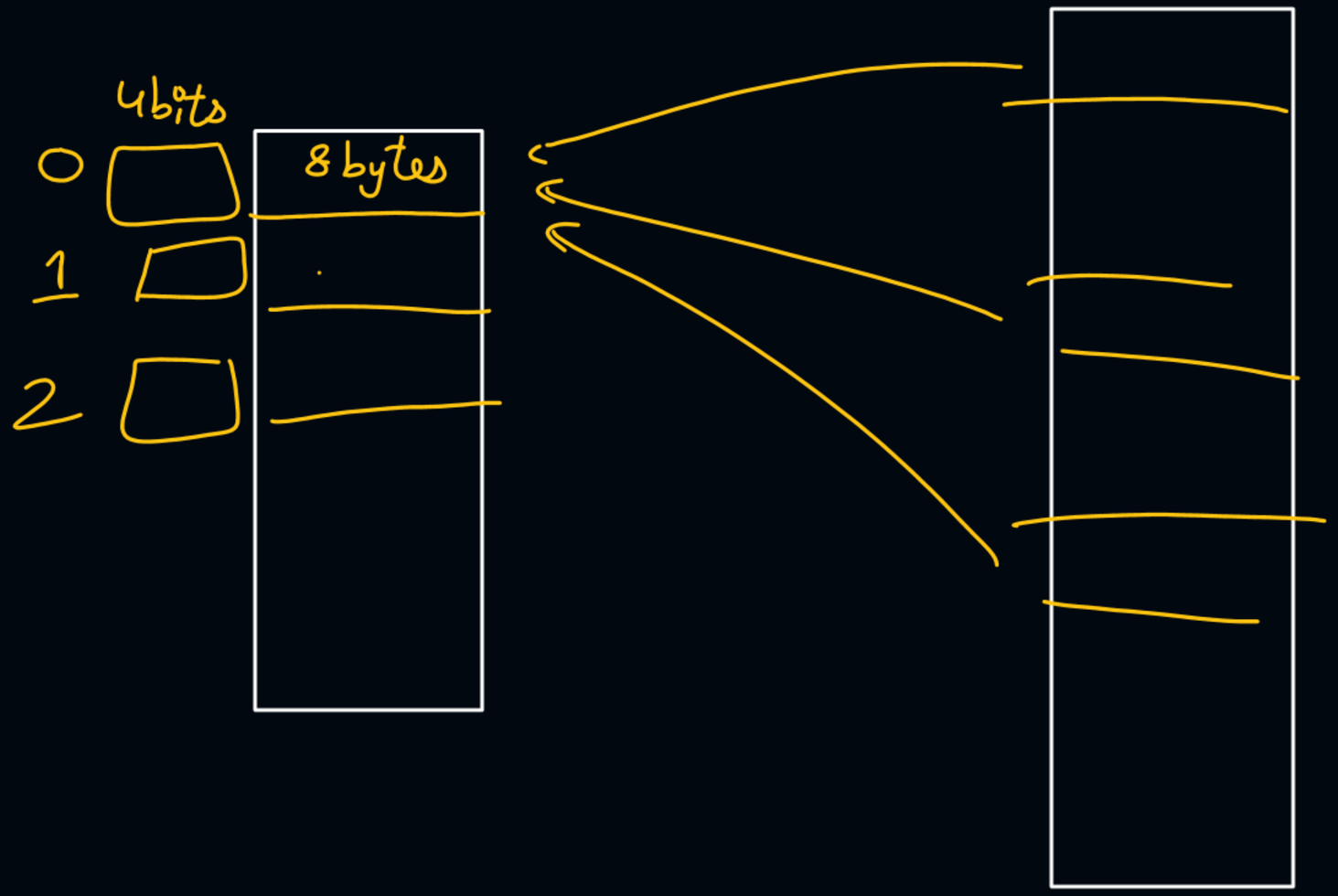
#Q. Consider a direct mapped cache of size 512B with block size 8 Bytes. The CPU generates 13-bits addresses.

1. Number of bits for byte offset? 3 bits
2. Number of blocks in cache? 2^6
3. The number of bits needed for cache indexing? 6 bits
4. The number of tag bits? 4 bits
5. Tag Directory size? $2^6 * 4$ bits = 256 bits



no. of blocks in cache = $\frac{512\,B}{8\,B} = \frac{2^9}{2^3} = 2^6$

CM block no. = 6 bits


4bits
8 bytes
0
1
2

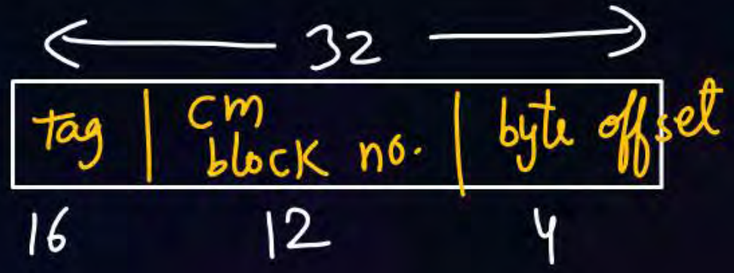
[NAT]

G. No. of bits for mm block no. = 28 bits

2^4 B ⇒ byte = 4 bits

#Q. Consider a direct mapped cache of size 64KB with block size 16 Bytes. The CPU generates 32-bits addresses.

1. Number of bits for byte offset? 4
2. Number of blocks in cache? 2^{12}
3. The number of bits needed for cache indexing? 12 bits
4. The number of tag bits? 16 bits
5. Tag Directory size? $2^{12} * 16$ bits = 64K bits = 8KB



no. of blocks in cache = $\frac{64KB}{16B} = 4K = 2^{12}$

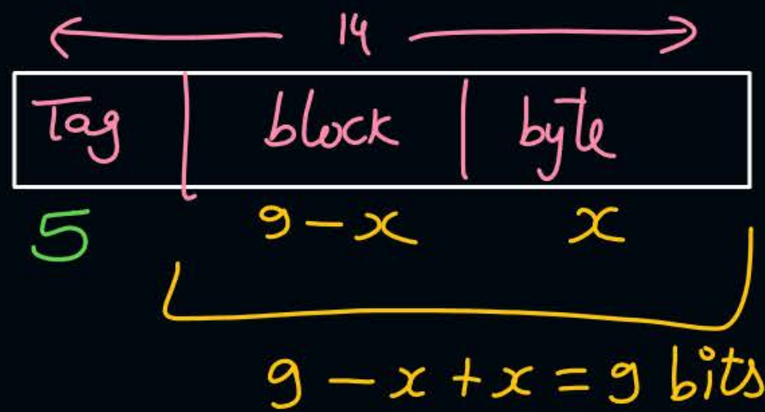
cm block no = 12 bits

Assume cm size = 512 B $= 2^9$ B

mm add. = 14 bits

block size = 2^x Bytes $\Rightarrow$ x bits for byte offset

Direct mapping



$9 - x + x = 9$ bits

$\log_2(512)$

$\log_2(\text{cm size})$

no. of blocks in cache $= \frac{512\ B}{2^x\ B}$

$= \frac{2^9\ \cancel{B}}{2^x\ \cancel{B}}$

$= 2^{9-x}$

$\Downarrow$

cm block no. = $(9 - x)$ bits

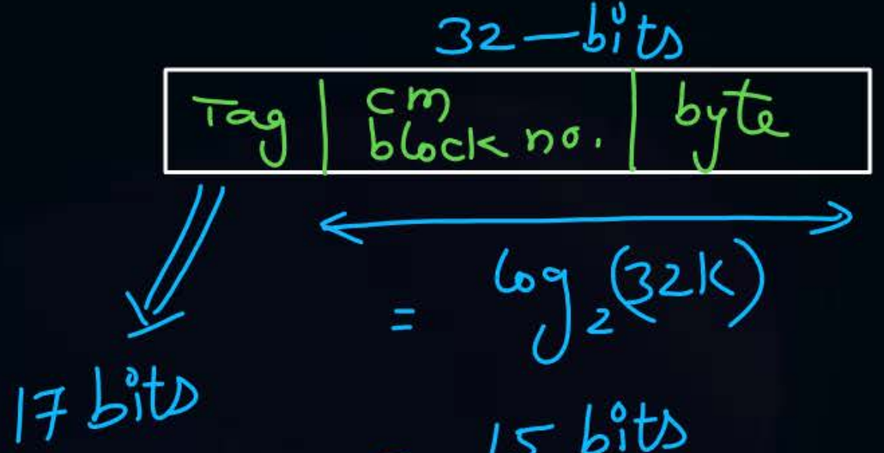
$$\text{Tag bits} = \text{mm add.} - \log_2(\text{cm size})$$

$$= \log_2(\text{mm size}) - \log_2(\text{cm size})$$

$$= \log_2\left(\frac{\text{mm size}}{\text{cm size}}\right)$$

[NAT]

#Q. Consider a direct mapped cache of size 32KB. The CPU generates 32-bits addresses. The number of tag bits in main memory address are?



Ans = 17 bits



Topic : Cache Initialization



Cache Memory

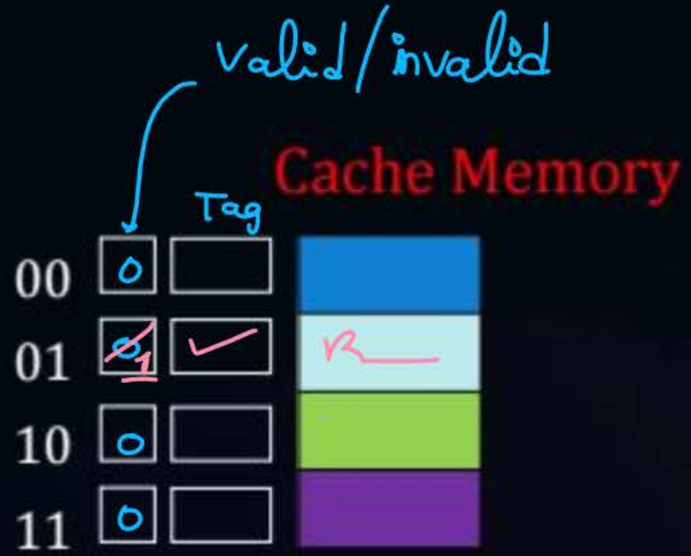


when computer turn on, cache is empty.

⇓

no any valid content in cache.

Topic : Cache Initialization



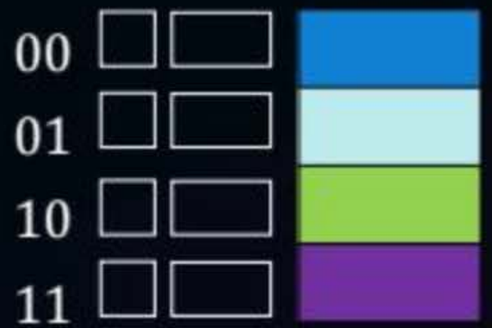
all valid/invalid bits are initialized by 0 when computer turn on.



Topic : Performance Improvement of Write Back Cache

while replacement write back only those blocks in which CPU performed any write.

Cache Memory



Topic : Performance Improvement of Write Back Cache

Dirty bits / modified bit → 0

→ 1

Valid/Invalid

Tag

Cache Memory



The replaced block has dirty bit 1, then only write it back to mm.

Tag directory size = no. of blocks in cache * (Tag + extra bits)

[NAT]

#Q. Block Size = 16 bytes

Size of Cache memory = 128KB

Size of Main memory address = 34-bits

Direct Mapping

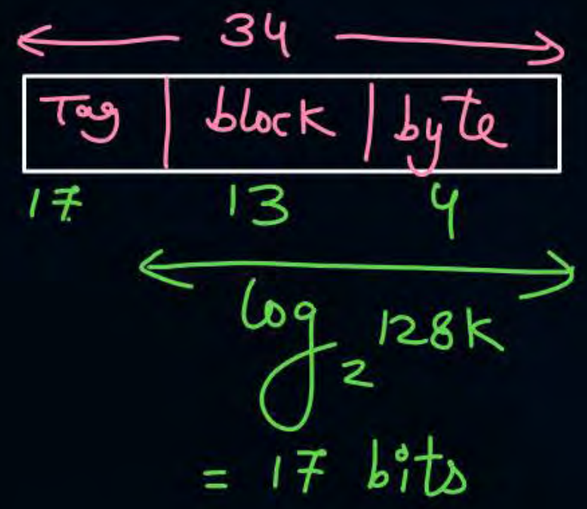
For each block apart from tag 1 valid bit and 1 modifies bit are stored in cache

Bits in byte offset? 4

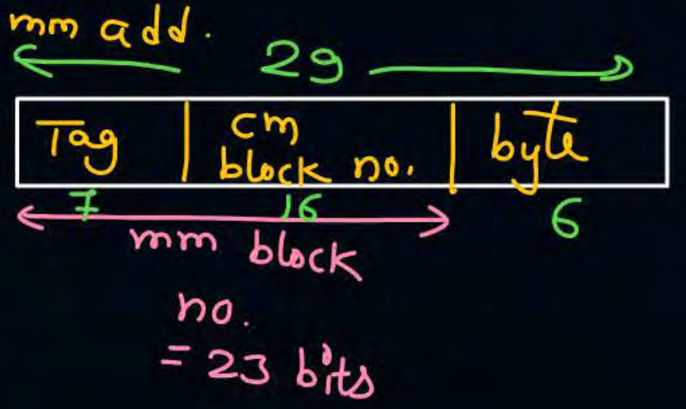
Bits in cache block number? 13

Bits in tag? 17

Tag Directory size? $2^{13} * (17+1+1) = 2^{13} * 19$ bits



[NAT]



#Q. Blocks in Main memory = 2^{23} ⇒ mm block no. = 23 bits

Blocks in Cache memory = 2^{16} ⇒ cm block no. = 16 bits

Block Size: 64 Bytes ⇒ 2^6 B

Direct Mapping

No. of bits required for Byte Offset = ? 6

No of bits required for main memory address = ? 29

Index-bits = ? 16

Tag-bits = ? 7

Size of Tag Directory = ? $2^{16} * 7$ bits

[NAT]

1 word = 32 bits = 4B

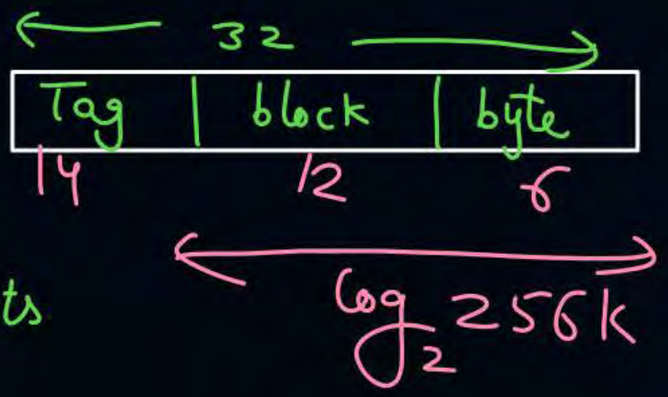
#Q. 32-bit architecture CPU

Main Memory Size = 4GB = 2^{32}B ⇒ mm add. = 32 bits

Cache Size = 256KB

Block Size = 16 Words = 16*4 = 64B = 2^{6}B

Direct Mapping



No. of bits required for Byte Offset = ? 6

No of bits required for main memory address = ? 32

No of bits required for main memory block no. = ? 26 bits

Index-bits = ? 12

Tag-bits = ? 14

Size of Tag Directory = ? 2^{12} * 14 bits

Topic : Calculating CM Block Number from MM Address

mm add.

→ 1. binary

→ 2. Hexadecimal

→ 3. Decimal

Tag	block	byte

ex :-

16

Tag	cm block no.	byte
6	7	3

mm. add.

1010111100010101

101011	1100010	101
Tag	cm block no.	byte offset

mm add :- C01B

1100 0000 0001 1011

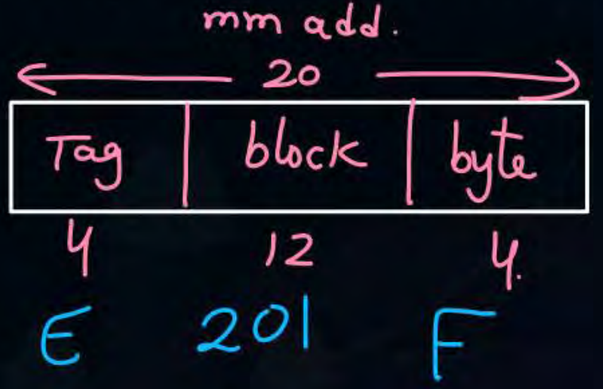
(30)16 ← Tag | cm block no. | Byte → (3)16

(03)16

[MCQ]

mm add. = 20 bits

#Q. Consider a machine with a byte addressable main memory of 2^{20} bytes, block size of 16 bytes and a direct mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be $(E201F)_{16}$ and $(E2020)_{16}$. What are the tag and cache line address (in hex) for main memory address $(E201F)_{16}$?



A ✓ E, 201

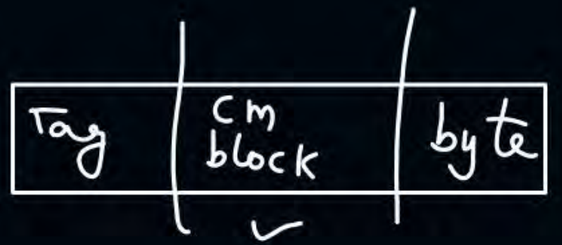
B F, 201

C E, E20

D 2, 01F

If mm add. given in decimal :-

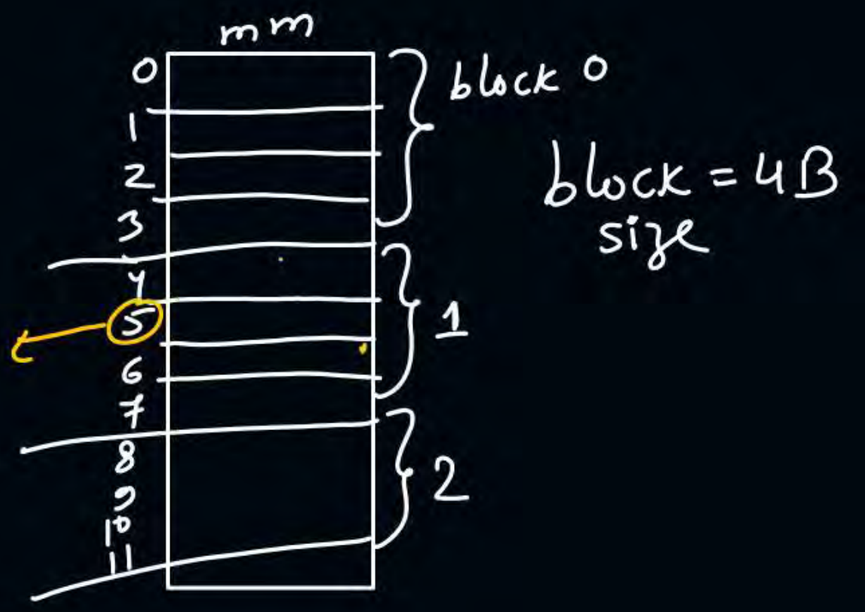
$$\text{mm block no.} = \left\lfloor \frac{\text{mm address}}{\text{block size}} \right\rfloor$$



$$\text{byte offset} = \text{mm add.} \% \text{ block size}$$

$$\text{cm block no.} = (\text{mm block no.}) \% \text{ no. of blocks in cache}$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of blocks in cache}} \right\rfloor$$



Topic : Checking Hit/Miss in Direct Mapped Cache



[MCQ]

H.W.

#Q. Consider a 64 bytes direct mapped cache with a block size of 16 bytes. Main memory size is 256bytes. Currently in the cache, the blocks are having tags as follows:

Block	Tag
00	10
01	01
10	11
11	01

Identify the correct statement with respect to the availability of the main memory data into cache?

a) Main memory byte number 243 present in cache
b) Main memory byte number 143 present in cache
c) Main memory byte number 43 present in cache
d) Main memory byte number 119 present in cache



2 mins Summary

Topic Cache Mapping

Topic Direct Mapping

Topic Tag

Topic Tag Directory



Happy Learning

THANK - YOU